

Affine Families and Inherited Structure in Unknot Recognition

Certified component optimization, normal-coordinate extraction,
and reuse of scalar endomorphism spaces

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ProveIt / Topology / UnknotRecognition

Mathematical development, implementation, and independent model-assisted review

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Abstract

We continue the ProveIt unknot-recognition program at a pinned repository revision. The main arithmetic result replaces enumeration by an exact optimization over an entire affine family. For a binary modulus m , a nonempty solution set $Az = b \pmod{m}$, and cyclic monodromies $h(z) = c + Hz$, we determine $\min_z \gcd(m, h_1(z), \dots, h_r(z))$ in deterministic polynomial bit time. The minimum is attained by a constructively produced assignment, without integer factorization. Compact dual modular identities certify both optimality and infeasibility independently of the optimizer. We connect this result to full-boundary assemblies of cyclic surface covers, allowing disconnected local covers and correlated seam parameters.

A second contribution extracts signed interval pairings and prism-gap contacts directly from admissible binary normal coordinates in tetrahedra, with a number of records linear in the triangulation size and independent of the number of normal discs. A third implements inheritance of scalar endomorphism spaces after verified chain-complex splitting, replacing descendant commutator solves by restriction of a transported parent basis. A dimension-saturated generator can additionally certify scalar locality and permit inheritance through one generator. The article gives proofs, exact-input contracts, independent finite checks, paired benchmarks, negative results, and a research agenda. These results improve specific represented operations. They do not prove a general quasi-polynomial unknot-recognition bound; global state growth, geometric admissibility, and hierarchy restart costs remain explicit.

Contents

1	The continuation and its exact scope	4
1.1	What problem is being improved?	4
1.2	The three bottlenecks addressed	4
1.3	Relation to the current geometric literature	5
1.4	Meaning of “proved” and of the reported novelty	5
2	Exact optimization over an affine family of cyclic monodromies	5
2.1	An affine coset contains a vector attaining its content ideal	6
2.2	Solving the seam equations with bounded residues	7
3	Independent certificates for optimality and infeasibility	8
4	From surface assemblies to affine loop families	9
4.1	The actual input model	9
4.2	A spanning tree removes redundant seam variables from the loops	10
4.3	A nontrivial connectedness example	11
4.4	Geometric and representation costs	11
5	What one affine optimum does not encode	12
5.1	Distinct framed states still exist	12
5.2	Connectedness does not determine genus	12
5.3	Separate component optima can conflict	12
5.4	Counting can be harder than producing an optimum	12
5.5	Many simultaneous peripheral conditions are hard	13
5.6	Completeness of a representation is a separate obligation	14
6	Extracting interval gluings from actual normal coordinates	14
6.1	The face-order calculation	14
6.2	Consecutive-disc prisms and their finite frontier	15
6.3	An explicit interface to interval orbit counting	17
6.4	Validation and measured scope	18
7	Inherited scalar algebras and conclusive local stopping	18
7.1	The full scalar commutant and its inherited corners	18
7.2	A conclusive scalar-locality certificate from dimension saturation	20
7.3	One-generator inheritance on saturated branches	21
7.4	Controlled families and precise performance claims	22

8	Global complexity: a precise remaining target	23
8.1	The family-query bound removes one parameter explosion	23
8.2	A composition theorem with every hypothesis exposed	23
8.3	Why a termination potential is insufficient	24
8.4	The two geometric interfaces must not be conflated	24
8.5	The algebraic route retains its intrinsic size barrier	24
9	Implementation, independent checks, and measured performance	25
9.1	Evidence and timing conventions	25
9.2	Arithmetic correctness and binary-size stress cases	25
9.3	Surface-family compilation and query boundaries	26
9.4	Normal-coordinate extraction: exact geometric and native audits	27
9.5	Attached copies: exact equation savings and a crossover	27
9.6	Field algebras: identify the work that actually matters	28
9.7	Natural scans: a useful negative timing result	29
9.8	Integrated regression and the inherited worker correction	30
10	Further research questions and prioritized experiments	30
10.1	From normal coordinates to a complete compressed cut	30
10.2	Stronger symbolic queries with controlled expressiveness	31
10.3	Scalar compression and actual scanner activation	31
10.4	A credible route to the global target	32
11	Reproducibility and repository integration	32
11.1	What the archive contains	32
11.2	Commands for independent reproduction	32
11.3	Concrete example contracts	33
11.4	Optional scalar controls and evidence boundaries	34
11.5	Provenance and continuing obligations	34

1 The continuation and its exact scope

1.1 What problem is being improved?

The long-term objective is a complete algorithm deciding whether a classical knot diagram with n crossings represents the unknot in time $n^{O(\log n)} = 2^{O((\log n)^2)}$. This is a statement about the entire computation, including preprocessing, state construction, every restart, coefficient bit lengths, and the final verdict. More generally, quasi-polynomial means $\exp((\log(n+2))^{O(1)})$; we focus on the concrete squared-logarithm target. A polynomial subroutine for an exponentially large state is not sufficient.

The maintained implementation combines inexpensive certificates and invariant filters with complete, potentially exponential fallback algorithms. Its Khovanov route uses a tangles-and-cobordisms formulation of chain complexes, following the framework of Bar-Natan [8]; the unknot-detection theorem itself is due to Kronheimer and Mrowka [9]. Exactness and the worst-case running time are separate questions. In particular, a filter which has not detected an obstruction has not proved the knot trivial.

This continuation audits the repository at `ea9cf443c8ce`, fetched on 8 October 2026. The initial remote inspection saw the immediately earlier revision `04d799b69d38`; the branch advanced while the sparse checkout was being obtained. All patches, baseline comparisons, and distributed source hashes use the actual checked-out revision `ea9cf443c8ce`. No inference is made from a mutable `main` URL after that point. The full baseline identifier is:

`ea9cf443c8ce33e09ed0c48b0ef7856ddf6326a3`

1.2 The three bottlenecks addressed

Recent incoming reports already supplied polynomial projectors, compressed surface-cover assembly, regular-cover gluing equivalence, and a binary normal-disc certificate checker [2, 3]. We use their precise scope, including their negative measurements. The present work advances three subsequent tasks.

1. **Optimize over a represented family.** Previous surface-cover kernels primarily classify a supplied assignment or compare two assignments. We ask an existential and optimization question over all assignments satisfying simultaneous linear congruences. This removes a genuine phase enumeration for the specified connectivity predicate.
2. **Extract compressed data from actual normal coordinates.** Rather than assume an already supplied sheet presentation, we construct signed disc-stack and prism-gap identifications from tetrahedral face pairings and seven normal coordinates per tetrahedron. This gives a concrete geometric interface, while stopping short of full compressed cutting and parallelity-bundle classification.
3. **Reuse algebra already computed.** The projector report identified the corner algebra $e\text{End}(C)e$ as the correct child endomorphism space but did not implement reuse. We implement and measure that inheritance, including canonicalization needed to preserve the existing bounded candidate policy.

Contribution	What is established	Remaining limitation
Affine cyclic families	Exact minimum component count; factor-free construction; dual certificates	Translation actions and affine constraints; no arbitrary topology predicate
Normal-coordinate extraction	Linear number of signed bands and local prism contacts; exact binary arithmetic	No full residual handle structure or unrestricted orbit implementation
Scalar-space inheritance	Exact child spaces from one parent solve; verified implementation	Parent dimension, component size, and splitter activation still matter

1.3 Relation to the current geometric literature

Lackenby’s July 2026 preprint [4] supplies an incompressibility algorithm and an unknot-recognition consequence. Its Section 9 does not provide cost bounds for every step; Proposition 9.1 instead bounds iterations by $L(g + 1)^L$. Theorem 14.4 already gives a polynomial-time compressed cutting operation for a uniform-type handle structure and a specified orientable normal surface, measured in handle count and logarithmic weight. Thus neither compressed cutting in principle nor the use of binary surface coordinates is a new theorem of this article. The new extraction code makes a smaller, explicit tetrahedral part of that interface executable.

Agol–Hass–Thurston’s interval-orbit methods [5] are the appropriate general background for processing enormous collections of normal pieces. Our local bounds are not a claimed improvement on their general orbit complexity. Covering-space monodromy and its connection with connected components are classical [6]. The affine-family theorem adds a fully specified optimization and certificate layer for a restricted family representation.

1.4 Meaning of “proved” and of the reported novelty

Theorems below have conventional mathematical proofs. The arithmetic, geometric, and algebraic implementations are supported by exact tests and independent finite oracles. Those are different forms of evidence: a finite test does not prove an asymptotic theorem, and a prose proof does not verify a Python implementation. Separate model-assisted reviews were used to challenge the proofs and check implementation contracts. No proof-assistant formalization or external peer review is claimed.

The delta relative to the inspected repository and incoming reports is explicit. We do not claim worldwide priority for the underlying facts about affine modules, primitive vectors, finite-ring duality, or corner algebras. Elementary ingredients can be useful even when they are classical; the research contribution here is the precise executable combination, its certificates, its performance consequences, and the identification of what it does and does not resolve.

2 Exact optimization over an affine family of cyclic monodromies

The input in this section is an explicitly supplied affine modular family. Let $m \geq 1$ be encoded in binary, let $R = \mathbb{Z}/m\mathbb{Z}$, and let

$$S = \{z \in R^n : Az = b\}, \quad h(z) = c + Hz \in R^\beta, \quad (1)$$

where $A \in R^{q \times n}$ and $H \in R^{\beta \times n}$. We charge the listed variable dimension n as part of the input size, even if every matrix has zero rows. More precisely, the size parameter is $n + q + \beta$ plus the binary length of m and all listed coefficients. This is the usual explicit-variable model, and it is also the appropriate output-sensitive measure for returning an n -entry witness. A short JSON integer declaring an enormous number of unused variables is not an encoding of those variables in this size convention. Initial coefficients are read and reduced modulo m before the bounded-residue arithmetic. For a connected base with translation monodromy generated by the entries of $h(z)$, the number of components is

$$\kappa(z) = \gcd(m, h_1(z), \dots, h_\beta(z)). \quad (2)$$

Indeed, every orbit is a coset of the subgroup generated by these translations, which is the subgroup of multiples of the displayed gcd. Representatives of the entries do not affect this gcd. With $\beta = 0$ the convention is $\gcd(m) = m$, as required for the trivial action on m sheets.

This is an optimization query about a supplied family, not a classification of all its marked members. Distinct assignments can have the same component count while retaining different attachment data. In particular, the theorem below does not eliminate an m^β lower bound for explicitly listing that many inequivalent marked states. When a pre-existing gluing model requires every block cover to be connected, global connectedness can already be forced by the assembly graph; the nontrivial use here allows disconnected local patches, or other affine families in which the monodromy subgroup varies.

2.1 An affine coset contains a vector attaining its content ideal

For a vector $v \in R^\beta$ and generator columns $u_1, \dots, u_k \in R^\beta$, define

$$\gamma = \gcd(m, \text{all entries of } v, u_1, \dots, u_k).$$

All gcds below are positive divisors of m .

Theorem 2.1 (Attained affine content). *There exist $t_1, \dots, t_k \in R$ such that*

$$\gcd\left(m, \text{all entries of } v + \sum_{i=1}^k t_i u_i\right) = \gamma.$$

Every vector in this affine coset has content divisible by γ . Consequently γ is its minimum content, both numerically and in the divisibility order. A witnessing coefficient tuple can be computed by a deterministic algorithm polynomial in the binary input length, without factoring m or enumerating parameter values.

A prime-by-prime existence argument would choose parameters modulo each prime power and invoke the Chinese remainder theorem. The following two-vector lemma supplies a direct factor-free implementation instead.

Lemma 2.2 (Factor-free content merge). *Let $a, b \in \mathbb{Z}^\beta$ represent two vectors modulo m , and put*

$$g = \gcd(m, a_1, \dots, a_\beta, b_1, \dots, b_\beta), \quad M = m/g, \quad d = \gcd(M, a_1/g, \dots, a_\beta/g).$$

If $d = 1$, take $t = 0$. Otherwise start $D = d$ and repeatedly replace

$$D \leftarrow \gcd(M, D^2) \quad (3)$$

until D is unchanged, and take $t = M/D$. Then $\gcd(m, a_1 + tb_1, \dots, a_\beta + tb_\beta) = g$. The saturation loop uses at most $1 + \lceil \log_2 \log_2 M \rceil$ gcd evaluations when $M \geq 2$.

Proof. Write $A = a/g$ and $B = b/g$, so that $\gcd(M, A_1, \dots, B_\beta) = 1$. For each prime p dividing M , write $e = v_p(M)$ and $f = v_p(d)$. After j saturation updates, the valuation in D is $\min(e, 2^j f)$. Thus at termination D contains the full prime-power factors of M supported on the primes of d , and no other primes. The integers D and $Q = M/D$ are coprime. This description proves the claimed iteration bound, since $e \leq \log_2 M$; it is a proof of the algorithm and does not ask the algorithm to find any prime factor.

If $p \mid D$, all entries of A vanish modulo p , some entry of B is nonzero modulo p , and Q is a unit modulo p . Hence $A + QB$ has a nonzero entry modulo p . If $p \nmid Q$, some entry of A is nonzero modulo p , while QB vanishes modulo p . In both cases the new vector has a nonzero entry. It therefore has content one modulo M , proving the stated content g modulo m . If $d = 1$, the vector A already has content one and no update is needed. $\square$

Proof of Theorem 2.1. Start with $a = v$ and merge successively with $u_1, \dots, u_k$ using Lemma 2.2. After the j th merge, the current vector has content equal to the gcd of m and all entries of $v, u_1, \dots, u_j$. Its coefficient on the previous vector is one, so the result always remains in the prescribed affine coset. Reducing entries and coefficients modulo m after each update bounds every stored residue by m . Products such as D^2 have at most twice this bit length; no unrestricted coefficient growth is hidden in the reduction. Each merge requires $O(\beta + \log \log(m + 2))$ gcd evaluations and $O(\beta)$ modular arithmetic operations on $O(\log(m + 2))$ -bit integers. Conversely γ divides every input entry and hence every affine combination. This proves attainment, minimality, and polynomial binary complexity. $\square$

The saturation is essential: for $m = 12$, $a = 2$, $b = 1$, the unsaturated choice $t = m/\gcd(m, a) = 6$ gives content four, whereas the saturated choice is $t = 3$, giving $a + tb = 5$ and content one. Testing just $t = 0, 1$ also fails already for $m = 6$, $a = 2$, $b = 1$.

2.2 Solving the seam equations with bounded residues

Classical polynomial algorithms compute integer Smith and Hermite normal forms together with transformation matrices [7]. For the present single-modulus problem, a simpler incremental reduction suffices and avoids reliance on a particular matrix package.

Maintain a complete parameterization of the equations processed so far,

$$z = z_0 + Bt \pmod{m}, \quad t \in R^n.$$

Initially $z_0 = 0$ and $B = I$. Zero or redundant columns may be retained; injectivity is unnecessary. For a new equation $a_0 z = b_0$, form

$$a = a_0 B \pmod{m}, \quad \delta = b_0 - a_0 z_0 \pmod{m}.$$

If $a = 0$, consistency is equivalent to $\delta = 0$. Otherwise swap a nonzero coefficient into the first position and reduce the row a to $(d, 0, \dots, 0)$ by extended-gcd two-column transformations. In one step, with current first coefficient d_0 and another coefficient a_j , choose $ud_0 + va_j = g = \gcd(d_0, a_j)$ and use

$$\begin{pmatrix} u & -a_j/g \\ v & d_0/g \end{pmatrix}.$$

Its determinant is one. It changes the parameter coordinates bijectively over R , and the corresponding two columns of B can be updated directly. All entries of B are reduced modulo m after each operation.

The remaining scalar equation is $dt_1 = \delta \pmod{m}$. Put $g = \gcd(d, m)$ and $Q = m/g$. There is a solution exactly when $g \mid \delta$. If so, choose

$$t_1^{(0)} = (\delta/g)(d/g)^{-1} \pmod{Q},$$

using $t_1^{(0)} = 0$ when $Q = 1$. Update z_0 by $t_1^{(0)}$ times the first column of B , and multiply that column by Q . This gives exactly all solutions of the enlarged system. The parameter dimension never grows beyond n .

Proposition 2.3 (Exact feasible-assignment count). *Starting with $N = m^n$, divide N by $Q = m/\gcd(m, a_1, \dots, a_n)$ at every feasible nonzero scalar restriction above. The final value is $|S|$, the number of distinct assignments satisfying the seam equations.*

Proof. The image of the new scalar functional on the previous solution module is the subgroup of R of size Q . Every nonempty fibre of a homomorphism between finite groups has the same cardinality. The new affine restriction therefore keeps exactly the fraction $1/Q$ of the previous solutions. Although the maintained parameterization can be redundant, all of its fibres over the previous solution module also have equal size, so the same fraction applies to distinct assignments. A zero consistent row has $Q = 1$ and makes no change. $\square$

After all seams have been solved, set

$$v = c + Hz_0, \quad U = HB \pmod{m}.$$

Theorem 2.1 gives the exact minimum of (2) and a witness $z_* = z_0 + Bt_*$. In particular, a connected member exists if and only if S is nonempty and

$$\gcd(m, \text{all entries of } v, U) = 1. \quad (4)$$

The constrained solver and the witness construction are both deterministic and polynomial in n, q, β and the binary coefficient lengths. For example, with classical arithmetic a conservative polynomial bound follows from $O(qn^2)$ modular matrix operations, $O(qn)$ extended gcds, $O(\beta n^2)$ operations for the image, and $O(n(\beta + \log \log(m+2)))$ gcds for the optimization. The final witness multiplication adds $O(n^2)$ modular operations. The auxiliary count $N \leq m^n$ has $O(n \log(m+2))$ bits, and its at most q exact divisions also have polynomial bit cost. This is not a claim of polynomial running time in the number of tetrahedra of an arbitrary knot exterior: constructing a suitable complete family remains a separate geometric problem.

3 Independent certificates for optimality and infeasibility

The optimizer need not be trusted to have generated a complete kernel basis. An exact optimum has a small certificate checked by elementary modular identities. Let $\mu \mid m$ be the proposed optimum and put $s = m/\mu$. Supply $z_* \in R^n$ and a matrix $Y \in R^{\beta \times q}$ satisfying

$$Az_* = b, \quad h_* = c + Hz_*, \quad \gcd(m, h_{*,1}, \dots, h_{*,\beta}) = \mu, \quad (5)$$

$$YA = sH, \quad Yb + sc = 0 \pmod{m}. \quad (6)$$

Theorem 3.1 (Primal and dual optimality certificate). *Identities (5)–(6) certify that S is nonempty and its exact minimum component count is μ . Every feasible input admits such a certificate for its true optimum. An infeasible input admits a vector $y \in R^q$ satisfying*

$$yA = 0, \quad yb \neq 0 \pmod{m}. \quad (7)$$

All these certificates can be generated in deterministic polynomial time without factoring m .

Proof. For every feasible z , the dual identities give $s(c + Hz) = sc + YAz = sc + Yb = 0 \pmod{m}$. Since $s = m/\mu$, each entry of $c + Hz$ is divisible by μ . Every component count is thus a multiple of μ , and the primal witness attains μ . Conversely, use the elementary annihilator identity

$$\{\ell \in R^{1 \times n} : \ell x = 0 \text{ for all } x \in \ker A\} = \{yA : y \in R^{1 \times q}\}. \quad (8)$$

Here is a proof valid also when m is composite. Write a Smith decomposition $PAQ = D$ with P, Q unimodular over $\mathbb{Z}$, hence invertible modulo m . For a nonzero diagonal entry d_i , put $g_i = \gcd(m, d_i)$. The corresponding kernel coordinate consists of multiples of m/g_i , so an annihilating row must have its i th entry divisible by g_i . That is exactly the ideal generated by d_i in R . A zero diagonal coordinate permits every kernel value and forces the row entry to be zero. Transforming back proves (8).

For the true optimum μ , Theorem 2.1 implies that every feasible holonomy vector has all entries divisible by μ . Each row of sH therefore annihilates $\ker A$, so (8) supplies the corresponding row of Y . Evaluating at any feasible z_0 also gives $Yb + sc = 0$. The rows of Y are found constructively by solving $A^T y_j^T = sH_j^T$ with the incremental congruence solver. When $\mu = 1$, one may simply take $Y = 0$.

For infeasibility, apply (8) to A^T : if b annihilated all of $\ker A^T$, it would lie in $\text{im } A$, contrary to infeasibility. Solve the homogeneous system $A^T y^T = 0$, then scan its generator columns for one whose dot product with b is nonzero. Such a column must occur. Equation (7) contradicts $Az = b$ directly. Every construction used here is polynomial. $\square$

The certificate verifier uses only modular matrix products and gcds; it does not invoke the congruence solver or the optimizer. The separately reported number of feasible assignments is exact by the preceding proposition, but is not part of this minimal certificate's contract.

4 From surface assemblies to affine loop families

4.1 The actual input model

Fix $m \geq 1$ and $k \geq 0$. Each listed patch is a compact connected surface F_v with nonempty boundary, described by orientability, genus or crosscap number, and boundary count. A canonical free generating system has $1 - \chi(F_v)$ generators. Every generator acts on the common fibre $\mathbb{Z}/m\mathbb{Z}$ by a translation

$$x \mapsto x + \alpha_{v,i}(z), \quad \alpha_{v,i}(z) = a_{v,i} + L_{v,i}z \pmod{m}.$$

All expressions use the same k parameters z . Local covers may be disconnected; this is essential for a nontrivial connectivity optimization.

A seam joins two different available boundary ports, possibly in the same patch. A port can be used at most once. Its base-circle degree is $\epsilon_e \in \{1, -1\}$, and its fibre map is $x \mapsto x + s_e(z)$,

where s_e is affine. The graph whose vertices are patches and whose edges are seams is required to be connected. Additional listed linear congruences on z are permitted. The input does not assert an embedding of these surfaces into a knot exterior.

For an orientable genus- g surface with b boundary components we use generators a_i, b_i and $c_1, \dots, c_{b-1}$; the final peripheral word is the inverse of the product of handle commutators and the c_i . For a nonorientable surface, squares of the crosscap generators replace the commutators. These words determine the peripheral translations $p_{v,j}(z)$ by signed addition. No arbitrary proposed peripheral system is trusted as a description of a surface.

Lemma 4.1 (Affine seam equations). *A seam from (u, i) to (v, j) with base degree ϵ_e and translation s_e is well-defined on the entire boundary preimage if and only if*

$$p_{u,i}(z) = \epsilon_e p_{v,j}(z) \pmod{m}.$$

Consequently all seam conditions and all supplied linear restrictions form one modular system $Az = b$.

Proof. Following a boundary loop before or after the seam gives respectively the translations $s_e + p_{u,i}$ and $\epsilon_e p_{v,j} + s_e$. Equality is precisely the displayed congruence. It is sufficient because an equivariant bijection of the fibre extends to an isomorphism of the corresponding boundary covers. Every expression is affine, so moving its constant term to the right produces a linear congruence. $\square$

4.2 A spanning tree removes redundant seam variables from the loops

Choose a rooted spanning tree of the patch graph. Set $t_o(z) = 0$ at the root and define t_v along tree edges by adding or subtracting the corresponding seam shifts. Thus t_v is the sheet displacement from the root fibre to the fibre in patch v . Store its affine coefficients reduced modulo m , together with its parent edge and direction.

The loop list consists of every patch generator translation and, for each non-tree seam $e : u \rightarrow v$, the affine displacement

$$t_u(z) + s_e(z) - t_v(z). \tag{9}$$

Conjugating a patch translation by the tree transport does not change it, since translations commute. The list therefore has

$$r = \sum_v (1 - \chi(F_v)) + |E| - |V| + 1$$

entries. Relations imposed by the surface gluings need not make this a free generating system; that is unnecessary for orbit computation.

Theorem 4.2 (Component count for a compiled family). *For every feasible z , let $h_1(z), \dots, h_r(z)$ be the compiled loop translations. The assembled covering surface has exactly*

$$\kappa(z) = \gcd(m, h_1(z), \dots, h_r(z)) \tag{10}$$

connected components, with the empty-list convention $\gcd(m) = m$.

Proof. Every point in the assembled cover can be connected to a point of the root fibre: first travel within its patch to the chosen fibre, then along the tree. Two points in the root fibre are connected precisely when a based loop in the assembled base sends one sheet to the other. Patch loops and

non-tree seam loops generate this action. Their image is the additive subgroup of $\mathbb{Z}/m\mathbb{Z}$ generated by the h_i . By Bezout's identity this subgroup consists exactly of the multiples of the displayed gcd. Its orbits are its cosets, and there are precisely that many. This proof also covers self-seams, the absence of non-tree seams, disconnected local covers, and $m = 1$. $\square$

Combining this theorem with the arithmetic optimizer gives a deterministic polynomial-bit-time procedure for the smallest possible component count over the entire feasible assembly family. The returned assignment is converted back to a concrete full-boundary assembly, using the input schema of the earlier surface-gluing kernel [2]. The certificate verifier recompiles the original surface data and checks the dual arithmetic identities; it ignores the producer's stored compilation. It therefore never accepts a modified loop list merely because it accompanies an otherwise valid arithmetic witness.

4.3 A nontrivial connectedness example

Take an annulus with twelve sheets, trivial local monodromy, and glue its two boundary preimages by a degree- -1 base map with fibre shift z . Before the gluing there are twelve disjoint annuli. After it there are $\gcd(12, z)$ components. The affine-family optimizer selects a unit shift and certifies a connected torus. This is a genuine change in global connectivity, not a reclassification of an already connected local block.

For a constrained example, give the annulus local monodromy 6 and impose $2z = 8 \pmod{12}$. The two feasible values are 4 and 10. The resulting component count is

$$\gcd(12, 6, z) = 2$$

for both. A single algebraic certificate proves that no connected member exists. Literal enumeration is unnecessary even when twelve is replaced by a binary integer with thousands of bits.

4.4 Geometric and representation costs

Let N count the listed patches, generators, seam endpoints, and constraint rows. Every affine expression that is present has $k + 1$ listed entries. We charge k , the listed dimensions, and all coefficient bit lengths as in Section 2; N need not count N nonempty expressions, since a disc patch can have none. Constructing the peripheral words and spanning tree, and adding their coefficient vectors, takes $O(N(k + 1))$ modular operations. The arithmetic solver charges its own matrix dimensions and coefficient bit lengths. The parameter count k is an explicit dimension, not a constant-cost binary instruction to construct a vector of astronomical length.

All loops remain translations. A negative base-boundary degree is not a reflection of the sheet fibre and is not an orientation character for the total surface. Allowing arbitrary signs or arbitrary affine units in the fibre would change the algebra and is outside this compiled model. The implementation preserves the base surface and seam directions so a later classifier can evaluate the realized surface's genus and orientability. Connectivity alone does not supply those answers.

5 What one affine optimum does not encode

5.1 Distinct framed states still exist

The previous regular-cover gluing report [3] counts m^β possible cyclic cycle-phase tuples after tree normalization. That count concerns distinct coverings over a fixed labelled decomposition. It cannot be removed when the requested output is an explicit list of every such state. Moreover, its local block covers are already connected, so a connected assembly of those blocks is connected for every seam phase. In that exact model, connectivity optimization would be trivial.

The current theorem concerns a different query on a broader input model: possibly disconnected local cyclic covers, with affine dependence in their monodromy and gluing data. It computes the optimum of one invariant over a family. It neither identifies all members with one another nor provides a complete invariant for all future attachments. This distinction prevents an incorrect inference from a polynomial existential query to a polynomial enumeration or canonicalization of all geometric states.

5.2 Connectedness does not determine genus

Example 5.1 (A five-sheet pair of pants). Let the base be a pair of pants, and let its first two boundary translations on $\mathbb{Z}/5\mathbb{Z}$ be 1 and a . The third is $-1 - a$. For $a = 1$ the cover is connected and has three boundary components, one over each base boundary. Since its Euler characteristic is -5 , its genus is two. For $a = -1$ it remains connected but has seven boundary components: one, one, and five. Its genus is zero.

The example follows directly from orbit lengths of a translation and $\chi(\tilde{F}) = 5\chi(F)$. It is also checked by the existing cover classifier in the regression suite. A hierarchy step that needs a planar base, a disc, a specific peripheral curve, or a particular attachment type therefore cannot substitute the component optimizer for that geometric test.

5.3 Separate component optima can conflict

Even a sum of simple component-count objectives can carry correlations. Over $\mathbb{Z}/2\mathbb{Z}$ consider two disconnected base graphs with respective single loop voltages z and $z + 1$. Each graph individually admits a connected cover. No common parameter makes both connected: their total component counts are $1 + 2$ or $2 + 1$. Thus optimizing each component separately and adding the answers would give the wrong result. The current surface-family compiler requires a connected base graph. A future multi-component interface must retain shared constraints and prove its joint optimization rule.

5.4 Counting can be harder than producing an optimum

For this subsection only, assume the prime divisors of m are known. Let S be the feasible parameter set, let $z_0 + Bu$ be a full parameterization, and write $h = v + Gu$, where $v = c + Hz_0$ and $G = HB$. For each prime $p \mid m$, let r_p be the rank of $G \pmod{p}$ and put

$$\theta_p = \begin{cases} 1, & -v \notin \text{im}(G \pmod{p}), \\ 1 - p^{-r_p}, & -v \in \text{im}(G \pmod{p}). \end{cases}$$

Proposition 5.2 (Exact proportion of connected assignments). *The number of distinct feasible parameters giving a connected cyclic cover is*

$$|S| \prod_{p|m} \theta_p. \quad (11)$$

Proof. Uniform $u \in (\mathbb{Z}/m\mathbb{Z})^k$ pushes forward to uniform $z \in S$, because every fibre of a homomorphism onto $S - z_0$ has the same cardinality. Modulo a prime p , the affine equation $v + Gu = 0$ has probability zero when inconsistent and probability p^{-r_p} otherwise. A vector is primitive modulo m exactly when it is not the zero vector modulo any prime dividing m . The Chinese remainder theorem makes these prime conditions independent for uniform u . Multiplying their probabilities and then by $|S|$ proves the formula. Redundant parameterizations do not alter the result. $\square$

This is a counting identity, not a factor-free algorithm for evaluating the product. The distinction is necessary. In the one-variable family $h(z) = z$ with no constraints, the connected assignments are precisely the units, so their number is Euler's totient $\varphi(m)$. If $m = pq$ for distinct unknown primes, knowledge of $\varphi(m)$ gives

$$p + q = m - \varphi(m) + 1,$$

and hence recovers p, q as the integer roots of a quadratic. Thus a general polynomial-time exact connected-assignment counter would imply a polynomial-time algorithm for factoring such semiprimes. This is a reduction, not a proof that the counting problem has no polynomial algorithm. It explains why the minimum and its witness can be obtained without offering an equally strong counting claim.

5.5 Many simultaneous peripheral conditions are hard

Theorem 5.3 (Simultaneous peripheral connectedness is NP-complete). *Consider the following decision problem: a connected planar bordered base surface is supplied with a three-sheet cyclic cover whose generator translations are affine functions of parameters in $\mathbb{Z}/3\mathbb{Z}$, and a subset of boundary circles is specified. Does some parameter assignment make the entire preimage of each specified boundary circle connected? This problem is NP-complete, even with no additional congruence constraints.*

Proof. Membership in NP follows by supplying the parameter vector. A boundary translation a has a connected three-sheet preimage exactly when $a \not\equiv 0 \pmod{3}$, so verification is polynomial in the listed input.

For hardness, take a finite graph $G = (V, E)$ and one parameter z_v for each vertex. Use a planar surface with $|E| + 1$ boundary circles and assign the first $|E|$ free generators the translations $z_u - z_v$, one for each edge uv . The final boundary has the inverse product, so its translation is minus the sum of these edge differences. Specify only the first $|E|$ boundaries. Their preimages are all connected exactly when $z_u \neq z_v$ for every edge, which is a proper three-colouring. Graph three-colourability is NP-complete [11]. The construction is polynomial even with dense coefficient rows. Restricting to graphs with at least one edge preserves hardness, and in every satisfying instance the entire surface cover is also connected, since at least one nonzero translation generates $\mathbb{Z}/3\mathbb{Z}$. $\square$

This is a hardness result for an expressive family query. It is not a hardness result for unknot recognition, normal-surface embedding, or a single already specified cover. It makes a practical

interface distinction: one global gcd objective is tractable, whereas imposing many correlated peripheral objectives permits graph colouring. Consequently a future hierarchy adapter must exploit geometric restrictions on its actual families rather than assume arbitrary affine peripheral constraints are equally inexpensive. The reduction also carries any quasi-polynomial algorithm for this unrestricted peripheral query to graph three-colouring. NP-completeness alone does not unconditionally rule out that running time; it identifies the strength of the additional algorithm being requested.

5.6 Completeness of a representation is a separate obligation

For a specified normal surface, the extracted interval maps are usually partial maps on several stacks of different sizes. They are not automatically translations of a single cyclic fibre. Likewise a normal surface search includes nonnegative coordinates and quadrilateral alternatives, not merely unrestricted residue parameters. Passing from those data to an affine cyclic family requires a separate extraction and coverage theorem. The two geometric interfaces in this article are useful complementary components; no unproved adapter between them is silently assumed.

6 Extracting interval gluings from actual normal coordinates

The input to this section is a finite tetrahedral triangulation $\mathcal{T}$ of a compact PL 3-manifold, with t tetrahedra, and an admissible normal vector $x \in \mathbb{N}^{7t}$. The manifold hypothesis is a mathematical precondition. The implemented extractor checks the face-pairing schema, reciprocal identifications, nonnegative coordinates, quadrilateral constraints, and matching equations; it does not check manifold links or a conversion from a knot diagram. Orientability of the ambient manifold is not needed for the extraction theorem.

Coordinates are ordered as

$$(T_0, T_1, T_2, T_3, Q_{01|23}, Q_{02|13}, Q_{03|12}).$$

A quadrilateral $Q_{ab|cd}$ avoids the opposite edges ab and cd . We distinguish the side containing vertex 0, denoted A , from its complement A^c . Triangle discs at vertex v are numbered from that vertex outwards. Quadrilateral discs are numbered from the side A towards A^c . All indices and intervals in the implementation are zero-based and half-open.

6.1 The face-order calculation

Consider the face opposite vertex f and an arc type cutting off vertex $v \neq f$. There is a triangle block of size T_v . There may also be a quadrilateral block: it is present exactly when the active quadrilateral separates $\{f, v\}$ from the other two vertices. In the order from v outwards, the triangle block precedes the quadrilateral block. If the latter has size Q , its j th disc has face rank

$$r(j) = \begin{cases} T_v + j, & f \in A, \\ T_v + Q - 1 - j, & f \notin A. \end{cases} \quad (12)$$

For triangles the face rank is simply $r(j) = j$.

This reversal is essential. One cannot number the quadrilateral stack outwards from all four face vertices simultaneously. To see the geometry directly, realize the quadrilateral discs by levels of

$\sum_{a \in A} z_a$ in barycentric coordinates. On a face missing a vertex of A , this sum is the coordinate of the cut-off vertex; on a face missing a vertex of A^c , the complementary sum is that coordinate. These orders are opposite.

A simplicial face pairing sends the corner v to its image corner and preserves the normal arcs' rank from that corner. Suppose source and target blocks use rank charts

$$r = \sigma i + \beta, \quad r = \tau j + \gamma, \quad \sigma, \tau \in \{1, -1\}.$$

On the intersection of their rank intervals, their disc indices satisfy

$$j = \tau \sigma i + \tau(\beta - \gamma). \quad (13)$$

Thus every nonempty block intersection produces an exact interval translation or reflection, with no individual discs constructed.

Theorem 6.1 (Linear normal-gluing extraction). *From $(\mathcal{T}, x)$ one can construct a polygon-gluing presentation of the represented normal surface using at most $5t$ nonempty polygon stacks and at most five affine interval pairings per paired tetrahedral face. Polygon templates specify the cyclic corner-edge labels, and every pairing retains its source face, target face, and vertex permutation. The presentation is normally isotopic to the surface specified by x .*

Let B be the largest binary coordinate length, with $B = 1$ for the zero vector. The construction uses $O(t)$ operations on integers of $O(B + \log(t + 2))$ bits. It therefore admits the conservative bit bound $O(t(B + \log(t + 2)))$ using binary addition, subtraction, and comparison, and has the same order of output size. In particular its complexity has no factor proportional to $\sum_i x_i$.

Proof. Each tetrahedron contains at most four triangle types and one quadrilateral type, giving the stack bound. Equation (12) gives their positions on each incident face. Their total arc counts agree across a pairing by the matching equations. Pair equal ranks and apply (13) on each overlap.

For the sharper face count, let $m \leq 3$ be the number of nonempty arc types on a paired face. On either side there are at most $m + 1$ blocks: every nonempty arc type contributes one, and only the type meeting the single active quadrilateral can contribute a second. Two interval partitions with a and b blocks have at most $a + b - 1$ nonempty overlaps. Summing over the m arc types gives at most $(m + 1) + (m + 1) - m = m + 2 \leq 5$ overlaps. If $m = 0$ there are none.

The polygon corners are labelled by the tetrahedron edges that they meet; the face permutation consequently specifies the endpoint identification of each paired polygon side. The normal discs can be simultaneously isotoped so that equal-ranked normal arcs coincide. Gluing these polygons therefore reconstructs the normal surface, including the incidences at triangulation edges, under the manifold precondition. Every loop of the construction has constant length per tetrahedron or paired face, and every arithmetic expression is a sum or difference of a constant number of coordinates. The stated time and size bounds follow. $\square$

6.2 Consecutive-disc prisms and their finite frontier

In one tetrahedron, the region between consecutive discs of the same type is a triangular or quadrilateral prism. A stack of $n > 0$ discs contributes $n - 1$ such local prism cells. The index i of a gap means the gap between discs i and $i + 1$.

Lemma 6.2 (The reflected-gap correction). *Suppose a surface pairing identifies disc i with $\varepsilon i + c$ on $[a, b)$, where $\varepsilon \in \{1, -1\}$. Its consecutive-disc prism pairing has domain $[a, b - 1)$, provided $b - a \geq 2$, and map*

$$i \mapsto \begin{cases} i + c, & \varepsilon = 1, \\ -i + c - 1, & \varepsilon = -1. \end{cases}$$

The sign ε records transport of the local interval fibre.

Proof. Both bounding discs must lie in the original pairing domain. The target gap is indexed by the smaller of the two target disc indices. For a reflection that smaller index is the image of $i + 1$, which is $-i + c - 1$. Omitting the final -1 gives the wrong prism. $\square$

Theorem 6.3 (Local residue and prism-frontier bounds). *Let $k_\Delta \leq 5$ be the number of nonempty disc types in a tetrahedron Δ , and let D_Δ be its number of normal discs. There are exactly $D_\Delta - k_\Delta$ between-disc prism cells and exactly $k_\Delta + 1 \leq 6$ other local complement cells. Their stack-end incidences can be written without expansion.*

Across a paired tetrahedral face, there are at most five compressed prism-pairing bands and at most two exceptional prism-to-residue contacts, counting both sides. Each exceptional contact is one prism side, not an interval of arbitrarily many such sides. For F_p paired faces and F_b boundary faces, all local prism sides can be classified using at most $5F_p$ prism-pairing bands and $2F_p + 4F_b$ frontier bands. Every frontier band either lies in the ambient boundary or specifies the opposite local residual cell.

Proof. The disjoint normal discs divide the tetrahedral ball into $D_\Delta + 1$ balls. For each nonempty type of multiplicity n , precisely $n - 1$ regions lie between consecutive parallel discs; the regions from different types are distinct. Subtraction gives $k_\Delta + 1$ residual cells. There is also a direct description. Without quadrilaterals there is one central cell and a vertex cap for each nonempty triangle type. With quadrilaterals there are two central cells, one on each side of the quadrilateral stack, and the same vertex caps. A triangle stack runs from its cap to the central cell on that vertex's side; the quadrilateral stack runs between the two central cells. These are the implemented labels and incidences.

Restrict each surface band by Lemma 6.2; this cannot increase the number of bands. A prism side fails to acquire a partner precisely when its two consecutive source arcs match target arcs of different types. The target face has at most one change of type: the triangle-to-quadrilateral transition in its quadrilateral's arc type. If that transition lies strictly inside a source block, it singles out exactly one source gap. Otherwise it yields no unpaired source prism. This gives at most one exceptional contact in each direction. The opposite region is the central residual cell on the side containing that arc's cut-off vertex, so its label is known.

A boundary tetrahedral face meets at most four stacks. Each contributes one interval of $n - 1$ boundary prism sides when $n \geq 2$. This proves the frontier bound and gives an exhaustive classification of every local prism side. $\square$

The constants five, six, and two are simultaneously attained. Glue face 0 of two tetrahedra by the identity permutation and use coordinate rows

$$(1, 1, 1, 1, 2, 0, 0), \quad (1, 2, 1, 1, 1, 0, 0).$$

Sharp face overlap: five gluing bands

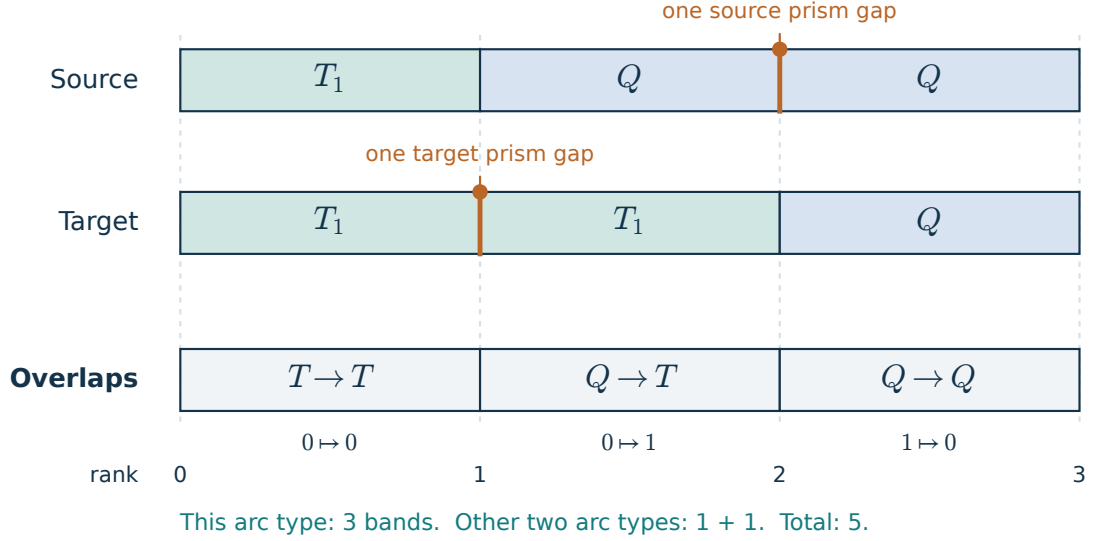


Figure 1: Sharp local extraction at cut vertex 1. The source blocks $[0, 1)$ and $[1, 3)$ and target blocks $[0, 2)$ and $[2, 3)$ yield three pairings. The two other arc types contribute one each, attaining five bands. Orange marks locate the two singleton prism gaps meeting opposite residual cells. The lower indices refer to the individual disc stacks.

For the arcs cutting off vertex 1, the two partitions are $[0, 1) \cup [1, 3)$ and $[0, 2) \cup [2, 3)$, giving three bands. The other two arc types give one band each. Both tetrahedra have six residual cells, and each direction has one exceptional prism contact.

These are local cut data. Around a triangulation edge, one must still assemble and regularize the incident pieces, reconstruct boundary-pattern attachments, and identify whole parallelity bundles. The output is not a claim that the residual-cell labels already form a valid global handle structure. In particular it is not an implementation of the full compressed-cut theorem in Lackenby’s hierarchy machinery.

6.3 An explicit interface to interval orbit counting

Let n_s be the multiplicity of stack s , let $o_s = \sum_{r < s} n_r$, and let $D = \sum_s n_s$. Concatenate the disc intervals into $[0, D)$. A pairing $i \mapsto \varepsilon i + c$ from stack s to stack u becomes

$$z \mapsto \varepsilon z + (o_u + c - \varepsilon o_s) \quad \text{on } [o_s + a, o_s + b).$$

This conversion is implemented as `orbit_input`. Its orbits are exactly the connected components of the normal surface: a path between polygon interiors can be moved away from the finite set of polygon vertices and then crosses a sequence of glued polygon sides. The algorithm of Agol, Hass, and Thurston [5] is the established polynomial-time method for counting such interval orbits; the new code prepares its input but does not implement that orbit-reduction algorithm.

The stored cyclic polygon orientations also give a signed version of each side pairing. If the pairing preserves the two induced boundary directions, consistent polygon-interior orientations differ

by a minus sign; if it reverses them, the interiors carry the same sign. The implementation prepares the orientation double cover by making two copies of every disc and switching copies exactly on negative transports. If C is the original orbit count and $\tilde{C}$ the count in this double cover, then

$$\#\{\text{orientable components}\} = \tilde{C} - C, \quad \#\{\text{nonorientable components}\} = 2C - \tilde{C}.$$

Each orientable component contributes two cover components and each nonorientable component contributes one. This is an aggregate count; obtaining the full coordinate vector and topology of each compressed component requires the weighted orbit machinery or an equally precise replacement.

6.4 Validation and measured scope

The exact geometric oracle realizes triangle slices by $z_v = 1 - (j + 1)/(4(n + 1))$ and quadrilateral slices by $\sum_{a \in A} z_a = 2/3 - (j + 1)/(3(n + 1))$. It explicitly sorts the resulting face arcs using rational numbers, independently of the block-overlay code. The suite covers all 24 tetrahedral vertex permutations, four source faces, and all choices of absent or active quadrilateral type on both sides, with fixed-seed small multiplicities. It includes 1537 cases, the sharp example above, and 591 reflected prism bands.

A separate comparison uses Regina [12] on 807 small normal-surface inputs. It compares full component coordinate multisets, Euler characteristics, orientability, and boundary counts. This includes 92 nonorientable and 274 closed components, scalar multiples of one-sided surfaces, and compatible normal sums that need not lie on extreme rays. Both interval-orbit adapters are also checked by explicit finite equivalence relations. Large-coordinate experiments run only the compressed extractor; neither the reference expansion nor Regina's component routine is invoked there.

For the previously supplied Fibonacci meridian family at $t = 128$, the input encodes about 2.79×10^{27} normal discs. The extractor returns 764 surface bands, 754 prism bands, 254 prism-to-residue contacts, and 512 local residual cells. At $t = 1024$ it returns 6140 surface bands and 4096 residual cells. Fixed-size tests with two tetrahedra and 65536-bit multiplicities retain ten stacks and four surface bands. These measurements demonstrate a representation and extraction improvement; they do not measure a complete unknot-recognition pipeline.

7 Inherited scalar algebras and conclusive local stopping

7.1 The full scalar commutant and its inherited corners

Work over $k = \mathbb{F}_2$. Partition the objects of a finite open complex C by their matching, homological degree, and recovered relative quantum shift. An allowed scalar change of basis is block diagonal for this partition. Write $b_1, \dots, b_t$ for the block sizes and

$$N = \sum_a b_a, \quad V = \sum_a b_a^2, \quad A = \text{End}_{\text{sc}}(C) \subseteq \bigoplus_a M_{b_a}(k).$$

The subscript is essential: these are coefficientwise commuting scalar endomorphisms. They need not be all endomorphisms in the cobordism category. Every morphism coefficient contributes its own homogeneous linear commutation equations. The maintained solver computes a complete basis of A whenever its configured variable cap permits a computation.

Theorem 7.1 (Full scalar corner inheritance). *Let S be an allowed invertible change of basis, and suppose the transformed complex has an actual direct-sum decomposition $S^{-1}CS = C_1 \oplus \cdots \oplus C_s$, with each C_i a whole connected component of its differential graph. Scalar blocks on a child carry the inherited parent labels; fresh relative quantum-shift recovery gives the same blocks on a connected child. Let e_i be the corresponding coordinate projection. Then*

$$\text{End}_{\text{sc}}(C_i) \cong e_i(S^{-1}AS)e_i.$$

In particular, restricting the transported members of any complete vector space basis of A spans the full scalar endomorphism algebra of C_i . This assertion does not require e_i to be central.

Proof. The coordinate inclusion j_i and projection p_i commute with the transformed differential, and $p_i j_i = 1_{C_i}$. Restriction sends a parent endomorphism a to $p_i S^{-1} a S j_i$. Conversely, if $u \in \text{End}_{\text{sc}}(C_i)$, then $j_i u p_i$, extended by zero on all other summands, is a typed scalar endomorphism of the transformed parent. Thus the restriction map is a surjective linear map, and its image is precisely the displayed corner algebra. Conjugating back by S proves the assertion about a basis. All changes preserve the parent quantum labels. On each connected child the recovered quantum labels differ from these inherited labels by a constant, so the permitted blocks agree. $\square$

A necessary algebraic distinction. The linear map $a \mapsto e_i a e_i$ from the whole parent algebra to its corner is not in general an algebra homomorphism. For example, in $M_2(k)$, with $e = E_{11}$, $a = E_{12}$, and $b = E_{21}$, one has $eabe = e$ while $eae = ebe = 0$. Multiplication on the corner itself is unambiguous; the algorithm needs the surjective linear map, not a quotient homomorphism assertion about the entire parent algebra.

Lemma 7.2 (Canonical-basis compatibility). *The existing nullspace solver eliminates equations with their most significant nonzero coordinates as pivots and lists its free variables in increasing order. Its output is exactly the unique reduced basis of the solution space whose pivots are least significant nonzero coordinates. Therefore reducing the restricted spanning vectors to this latter form produces the same ordered basis as a fresh child commutation solve.*

Proof. Each nullspace basis vector has one free coordinate equal to one and all other free coordinates zero. The equation pivots determined by that free coordinate occur only at larger indices. Hence the free coordinate is its least significant nonzero coordinate, and no such pivot coordinate occurs in another basis vector. These properties characterize reduced row-echelon form for the reversed choice of pivot direction and determine the basis uniquely. $\square$

This compatibility matters operationally: the bounded basis-prefix search and the fixed-seed random combinations then inspect exactly the same candidates, in the same order. General corner inheritance changes the construction of the search space, not the search policy. With exact arithmetic and absent an intervening resource interruption, it preserves the accepted split witnesses and the resulting recursive component order.

Computational scope. On a cold recursion tree, general inheritance computes the full scalar commutant once at the root and derives every subsequently examined child space by linear algebra, without constructing another commutator system. An accepted split loaded from an older cache entry need not have a retained commutant basis; ordinary inheritance can then resume at a descendant. The cache retains the previous witness format rather than storing large algebra bases indefinitely.

The implementation has two exact transport kernels. For a low-dimensional algebra it directly forms $S^{-1}aS$ for its few basis matrices. For a larger algebra it constructs one packed linear map on matrix variables. The image of a matrix unit is obtained from

$$S^{-1}E_{uv}S = (S^{-1}e_u)(e_v^T S),$$

then restricted to all selected children. One application of this packed linear map transports each parent basis vector. Both kernels give identical canonical child bases. Their selection affects performance only.

For reference, direct dense transport of a D -dimensional algebra costs $O(D \sum_a b_a^3)$ elementary field operations before child basis reduction. If child i has V_i scalar variables, reducing D projected generators costs at most $O(DV_i \min(D, V_i))$ elementary field operations; packed binary arithmetic improves the corresponding word-operation bound. These bounds do not imply that transport always beats resolving a sparse child commutator. The saved quantity is the number of commutator-system constructions and solves, while the replacement algebra computations have their own cost. Moreover, cache keys, grading recovery, accepted-witness verification, and transformation of the differential still inspect morphism data; the whole recursive scanner is not coefficient independent.

7.2 A conclusive scalar-locality certificate from dimension saturation

Theorem 7.3 (Dimension-saturated scalar locality). *Let A be the complete scalar endomorphism algebra, of dimension D , and let $Q \in A$ have actual minimal polynomial $m_Q(t)$. If*

$$\deg m_Q = D,$$

then $A = k[Q] \cong k[t]/(m_Q)$. If in addition

$$\dim_k \{p \in k[t]/(m_Q) : p^2 = p\} = 1,$$

then C has no nontrivial scalar direct-sum decomposition.

Proof. The elements $1, Q, \dots, Q^{D-1}$ are independent by the definition of the minimal polynomial. They lie in the D -dimensional space A and hence form a basis. Evaluation therefore identifies the whole algebra A with the displayed polynomial quotient. In characteristic two, squaring is linear on this commutative quotient. A one-dimensional fixed space contains exactly 0 and 1, so these are its only idempotents. A nontrivial scalar direct sum would supply a nonzero nonidentity scalar coordinate idempotent, a contradiction. $\square$

To see the equivalent factorization criterion, write $m_Q = \prod_{j=1}^r f_j^{e_j}$ with distinct irreducibles f_j . In a primary quotient $k[t]/(f_j^{e_j})$, an idempotent u satisfies $f_j^{e_j} \mid u(u-1)$. Since u and $u-1$ are coprime, the full power divides one factor, giving $u = 0$ or 1 in the quotient. The Chinese remainder theorem therefore gives exactly 2^r idempotents, and the Frobenius fixed space over $\mathbb{F}_2$ has dimension r . Thus fixed-space dimension one means that m_Q has one distinct irreducible factor and the quotient is local, including repeated factors. This argument explains the criterion; the implementation obtains the fixed space by linear algebra, using the inherited Berlekamp routine [10, 2], without requiring an integer factorization. A certificate does not require that the algebra be a field. The tests include both $\mathbb{F}_8$ and $\mathbb{F}_2[t]/((t^2 + t + 1)^2)$, exhaustively checking that their displayed commutants have exactly two idempotents.

Both hypotheses are indispensable operationally. The dimension must be that of the complete commutant, and the polynomial must be the actual minimal polynomial, not merely an annihilating polynomial. A primary candidate in a proper subalgebra proves nothing about idempotents elsewhere in A . For example, the identity of a decomposable complex has primary minimal polynomial $t + 1$ while its full commutant has dimension greater than one. An explicit negative control verifies that this case is rejected.

The implementation reuses the earlier polynomial-projector routine’s exact minimal polynomial and Berlekamp fixed-space computation. When a projector search fails and its minimal degree equals the full commutant dimension, the optional locality flag terminates the otherwise bounded candidate search with a certificate. This shortcut cannot suppress a possible scalar split: the certificate establishes that none exists. It makes no assertion about non-scalar cobordism endomorphisms.

The separate verifier recomputes the entire commutant, verifies commutation of the proposed generator, recomputes its minimal polynomial, and checks the fixed-space dimension. It deliberately incurs the full cost again; certificate auditing is outside benchmark timing. Recorded certificates include all differential rows, matchings, frontier points, and degrees, and survive JSON round trips.

7.3 One-generator inheritance on saturated branches

Corollary 7.4 (Inheritance from one saturated generator). *Under the child-label hypotheses of Theorem 7.1, suppose $A = k[Q]$ has been certified by dimension saturation. For any actual child coordinate idempotent e_i after an allowed change of basis, its full endomorphism algebra is generated by the restricted Q_i and its own identity:*

$$\text{End}_{\text{sc}}(C_i) = k[Q_i].$$

Consequently its dimension is $\deg m_{Q_i}$; if m_{Q_i} is primary, the child is scalar indecomposable without any new commutator solve.

Proof. Conjugation preserves the polynomial-algebra identity. The parent algebra is commutative, so its child selectors are central. Its corner consists of the elements $e_i p(Q) e_i = p(Q_i)$, with constants interpreted as multiples of the corner identity. The corner theorem identifies this with the full child algebra. Minimal-polynomial dimension and the saturation theorem now apply. $\square$

Thus only one generator needs to be transported, rather than a full parent basis. The implemented optional path recognizes saturation on an accepted polynomial-projector witness, transports its original generator, and checks each examined child’s minimal polynomial. A local child is finished immediately. For a nonlocal child, the powers of the restricted generator are reduced to the same canonical commutant basis as an independent solve, and the existing bounded splitter resumes.

There is a stronger algorithmic consequence, which should be distinguished from the implementation just described. Factoring m_Q into its primary factors, or recursively using the available polynomial projectors, yields a complete scalar direct-sum decomposition of a saturated monogenic algebra when split budgets permit. The current implementation completely finishes the measured two-field branches; it does not replace every nonlocal child’s bounded search by an unlimited recursive factor-decomposition procedure.

7.4 Controlled families and precise performance claims

For the existing attached-copy family with $r \geq 2$ copies, the commutant is $M_r(k)$, there are $3r^2$ scalar variables, and each size- r solve builds $2r^2$ nonzero coefficient equations. With sufficiently large split budget, the original deterministic policy splits off one copy at each step. Its total equation construction count is therefore

$$2 \sum_{j=2}^r j^2 = \frac{r(r+1)(2r+1)}{3} - 2,$$

whereas inheritance builds exactly $2r^2$ such equations at the root. This is an exact cubic-to-quadratic reduction in the number of constructed commutator equations, not a claim of a corresponding total runtime bound. The basis candidates and accepted witnesses remain identical. Both the formula and the witness identity are regression tested through $r = 16$.

The separately inherited two-field fixtures have a homogeneous two-term differential $d = xI + yB$ over a two-dot algebra, where B is conjugate to companion matrices for two distinct irreducibles of degree r . Comparing the independent x and y coefficients shows that a scalar endomorphism is a common matrix commuting with B . Thus

$$\text{End}_{\text{sc}}(C) \cong k[t]/(fg) \cong \mathbb{F}_{2^r} \times \mathbb{F}_{2^r}.$$

These are legitimate graded algebraic complexes, without a claimed realization as knot-scan prefixes. Their root has $8r^2$ scalar variables. For the recorded dense fixtures at $r = 8, 10, 12$, the accepted generator saturates the algebra. One root commutant solve, two candidate tests, and two inherited scalar-locality certificates complete the decomposition. The earlier primary policy used respectively 46, 52, 58 candidates over the whole recursive compression and three commutant computations.

Natural-knot scans, attached-copy tests, and field-algebra tests have separate timing scopes. The benchmark retains randomized paired samples and A/A controls. It compares general inheritance alone, local stopping alone, their ordinary combination, and one-generator inheritance independently. No controlled-family speedup establishes a worst-case bound for arbitrary knot diagrams. In particular, nothing here bounds the number of objects, matching types, algebra variables, or scalar-indecomposable summands of a general scan by a quasipolynomial function of its crossing number.

Provenance and proposed follow-up questions. The polynomial-projector module and the two-field fixtures are reproduced unchanged from the earlier October 8 projectors package [2], with their hashes recorded separately. Their algorithm is prior work, not a new contribution of this continuation. The additions are complete-corner transport, canonical basis preservation, dimension-saturation stopping and verification, and one-generator inheritance on certified saturated branches.

The next questions are whether to replace the residual bounded scalar search by constructive finite-algebra radical and semisimple-quotient algorithms; whether an exact cost model can select transport versus resolving a sparse commutator; whether additional natural knot families produce saturated or otherwise tractable scalar algebras; and which structural bounds control the number and size of scalar-indecomposable components. Extending reuse across the addition of a crossing is a different problem: a child corner of the same complex inherits all endomorphisms by extension by zero, while a newly glued complex can gain endomorphisms that have no ancestor.

8 Global complexity: a precise remaining target

8.1 The family-query bound removes one parameter explosion

Suppose a represented stage previously tried up to m^k phase vectors, and the only information requested from that family is feasibility and the minimum number of cyclic covering components. For the affine model of this article, compilation and optimization have polynomial bit cost in the listed matrix dimensions and input integer lengths. Thus the m^k term can genuinely be removed from that stage. The emitted witness and its independent verification are polynomial as well. This conclusion does not require $k = O(\log n)$ or m to be a small integer.

There is no corresponding conclusion for a stage which must distinguish all seam assignments under future attachments, require many selected boundary preimages to be connected, or test unknown normal-surface embeddings. Sections 4 and 5 give examples and formal barriers to those extensions.

8.2 A composition theorem with every hypothesis exposed

Theorem 8.1 (Conditional represented-algorithm bound). *Consider a complete unknot-recognition algorithm on n -crossing diagrams. Suppose that, on every execution:*

1. *the number of represented states and transitions, including all failed branches and resets, is at most $\exp(C \log^2(n + 2))$;*
2. *every listed state and certificate has bit size at most $\exp(C \log^2(n + 2))$;*
3. *every transition is either a verified operation proved polynomial in that bit size, or another routine whose total cost has the same quasi-polynomial bound;*
4. *the representation and transition relation are sound and complete for the required topological decision.*

Then the total bit time, including certificate checks and output, is $\exp(O(\log^2(n + 2)))$.

Proof. A fixed polynomial in $\exp(C \log^2(n + 2))$ is $\exp(O(\log^2(n + 2)))$. Multiplying this by the assumed bound on the number of transitions preserves the form. Reading and emitting every state is included in the same product. The soundness and completeness hypothesis is needed to identify the resulting algorithm as a complete recognizer rather than an exact subroutine or a partial search. $\square$

This bookkeeping theorem is elementary. Its usefulness is diagnostic: the arithmetic optimizer and normal-coordinate extractor establish particular transition-cost statements in item (3), under their explicit input contracts. Scalar-space inheritance avoids repeated work on an already represented complex. None establishes items (1), (2), or (4) for unrestricted knots. In particular, the theorem is not presented as a new solution to the central complexity problem.

8.3 Why a termination potential is insufficient

A lexicographically decreasing list with L entries each in $\{0, \dots, g\}$ can visit $(g+1)^L$ values. The corresponding integer potential is

$$P(a_1, \dots, a_L) = \sum_{i=1}^L a_i (g+1)^{L-i}.$$

Decreasing an earlier coordinate and resetting all later coordinates to g decreases P , but can make progress of only one unit. This is exactly the type of accounting behind the hierarchy iteration bound $L(g+1)^L$ in Lackenby’s Proposition 9.1 [4].

A sufficient numerical regime for this particular bound is

$$\log(L+1) + L \log(g+1) = O(\log^2(n+2)),$$

together with control of the per-step bit cost. The first term also covers $g=0$, when the bound reduces to L . A merely linear estimate for L does not supply it. Faster gcds, shorter certificates, or reduced commutant solves do not change the number of possible resets. A new global argument would need, for example, a stronger amortized progress measure, a decomposition theorem limiting genuinely new states, or an operation that processes a whole relevant reset family symbolically.

8.4 The two geometric interfaces must not be conflated

The normal extractor produces partial signed interval maps, with source and target stacks and explicit endpoint data. The family optimizer receives total translations of a common cyclic fibre and affine parameter constraints. These are different mathematical objects. A sound bridge could identify a cyclic block inside the interval data, certify that all its possible completions are represented, and pass the induced congruences to the optimizer. No such general bridge is included.

Likewise, computing the local complement cells is less than constructing their full residual handle structure, including every attachment and boundary pattern. General orbit counting can determine connected pieces, but subsequent genus, orientation, and vertical-annulus data must retain the appropriate provenance. Theorem 14.4 of the geometric literature identifies the desired full uniform-type handle-structure interface [4]; the present implementation supplies a concrete first portion of it.

8.5 The algebraic route retains its intrinsic size barrier

Transporting an endomorphism space can avoid repeatedly solving for the same information. A certified local scalar endomorphism algebra can stop a futile search for scalar direct summands. Neither fact bounds the number of surviving homological generators or the size of the parent commutant. A complex with a local endomorphism algebra can still be large. The first computation of that algebra may dominate, and a bounded search can remain inconclusive when no dimension-saturated generator is found.

The measured implementation therefore retains caps, transactional failure behavior, and separate optional controls. A local algebra certificate means “no nontrivial splitting by the admitted scalar maps.” It does not mean “unknot,” “small homology,” or “no more general continuation-compatible compression.”

9 Implementation, independent checks, and measured performance

9.1 Evidence and timing conventions

The proofs establish asymptotic claims for their stated input models. Independent finite checks test whether the implementation realizes those models. Timings answer a narrower question about the cost of these particular implementations in the recorded environment. These three forms of evidence are reported separately.

The recorded runtime was Python 3.12.14 on Linux x86_64, kernel 6.18.44, with glibc 2.39. The optional native geometry audit used the Regina 7.4.1 Python distribution; its library version string reports 7.4. Computations ran in a shared environment without CPU isolation. Absolute elapsed times should therefore be treated as descriptive measurements, not portable performance predictions. In particular, repeated geometry runs varied under shared load. The final recorded JSON files, rather than earlier working measurements, supply every table below.

The arithmetic measurements use three repetitions per input and include certificate generation and the optimizer’s independent certificate check. Normal extraction uses five repetitions on the layered-torus sequence and seven for fixed triangulation size. Fixtures are built before these timed regions. No expanded surface or native component computation is called on the large normal-coordinate cases.

The scalar benchmark uses seven randomized paired rounds. Each arm has a fresh state, an untimed warm-up, and between one and five calls per round, with batch size recorded in the data. A/A controls run the reference twice. A reported ratio is

$$\text{median}_{j=1}^7 \frac{T_{\text{reference},j}}{T_{\text{variant},j}},$$

so values greater than one favor the variant. This differs in general from the ratio of the two separately reported median times. Exact output comparisons and scalar-locality certificate audits are outside these timed regions. No confidence interval or natural-knot speedup is inferred from a few percent of timing variation.

9.2 Arithmetic correctness and binary-size stress cases

The standalone arithmetic tests use exhaustive enumeration of the original variables as the reference, not the parameterization returned by the new solver. The exhaustive and seeded comparisons total:

Independent arithmetic comparison	Cases
Scalar merges, $m \leq 48$	38 024
Two-coordinate merges, $m \leq 8$	8 772
Two-generator scalar cosets, $m \leq 10$	3 025
Scalar congruences, $m \leq 24$	4 900
2×2 systems, $m \leq 4$	4 890
One-variable one-seam families, $m \leq 6$	2 275
Seeded dense random families	2 000
Total	63 886

Additional tests cover empty parameter and holonomy dimensions, modulus one, cooperative cancellation, malformed input, and deliberately corrupted certificates. The corresponding thirteen

`unittest` methods are included in the integrated suite. The verifier can be run without calling the optimizer or trusting a supplied kernel basis.

Family	Modulus bits	$n/q/\beta$	Min.	Seconds
$2 + z$, no seams	2 648	1/0/1	1	0.000081
$2 + z$, no seams	42 353	1/0/1	1	0.008106
$2 + z$, no seams	169 409	1/0/1	1	0.191455
Dense constrained	131	24/12/7	6	0.019229
Dense constrained	515	24/12/7	6	0.039649
Dense constrained	2 051	24/12/7	6	0.231718
Dense constrained	8 195	24/12/7	6	2.446944

Here $n/q/\beta$ gives the variable, constraint, and holonomy dimensions. For the last one-parameter row, $m = 6^{65536}$ has 169,409 bits. The returned witness is $z = 3^{65536}$, and exactly seventeen saturation gcds are used. The algorithm represents no individual sheets and enumerates no parameter tuples. Its work is controlled by binary length, not by the astronomical integer value of m .

The dense stress families have a deliberately planted exact optimum six. For a seeded dense matrix A , a feasible hidden vector has every coordinate divisible by six. The first six holonomy rows are $A_j + 6R_j$, and the last holonomy is the constant six. On the feasible locus all coordinates are multiples of six, and the last coordinate forces their gcd with m to be six. The returned certificates have six nonzero dual entries. This design provides a known independent answer while exercising dense congruence elimination and nontrivial dual certificates; it is not presented as a sample of difficult knot exteriors.

9.3 Surface-family compilation and query boundaries

The geometric wrapper has nineteen regression methods. Its seeded test constructs 120 families, of which 51 are feasible and 69 are infeasible. It examines 1,337 original parameter assignments: 437 satisfy the seams, and 900 fail them. For each feasible assignment, a direct disjoint-set calculation on patch-sheet pairs agrees with the compiled component count. This reference does not use spanning-tree potentials or the compiled loop list. Every family’s minimum and feasible-assignment count agree with the literal results, and both kinds of certificate verify.

Targeted cases check a tree edge traversed backwards, reversal of a seam, nonorientable peripheral squares, both base-boundary degrees, and gluing that connects locally disconnected covers. The emitted large-integer JSON certificates also survive a serialize–parse–verify round trip. This last check is material: a verifier that accepts a large integer in memory but rejects its documented hexadecimal transport is not a usable artifact verifier. The wrapper decodes only documented certificate integer fields; it does not interpret unrelated metadata as mathematics.

A separate check of Theorem 5.3 uses the maintained surface-cover classifier to inspect the full lifted-boundary profiles for all 351 assignments of three graph families:

Graph	Assignments	Globally connected	All designated boundaries connected
K_3	27	24	6
K_4	81	78	0
C_5	243	240	30

These checks are distinct from the 63,886 arithmetic cases. The K_4 example makes the interface boundary concrete: a family can have many globally connected members and no member satisfying all requested peripheral connectedness conditions.

9.4 Normal-coordinate extraction: exact geometric and native audits

The local geometric oracle realizes triangle and quadrilateral discs by rational barycentric levels and sorts the actual face intersections. Its 1,537 cases cover all 24 vertex permutations, all four source faces, and all absent/active quadrilateral combinations on each side. The two methods agree on 9,771 individual disc identifications and 4,850 prism-gap identifications. The latter include 591 reflected bands, which exercise the otherwise easy-to-miss -1 in the reflected gap map.

A separate Regina comparison covers 807 small normal-surface inputs and 1,537 component coordinate/topology records, including 92 nonorientable and 274 closed components. It checks full component-coordinate multisets, Euler characteristics, orientability, and boundary counts. Scalar multiples of one-sided surfaces and compatible nonextreme sums are included. Eight additional package tests exercise the command-line interface, callbacks, schema errors, matching failures, empty surfaces, and binary coordinate transport. These finite comparisons support the implementation; they do not replace the extraction proof.

For the Fibonacci meridian family inherited from the earlier certificate report, the final measurements are:

t	Coordinate bits	Surface	Prism	Contacts	Residue	ms
8	108	44	34	14	32	0.833
16	342	92	82	30	64	1.679
32	1 214	188	178	62	128	3.533
64	4 554	380	370	126	256	7.162
128	17 635	764	754	254	512	15.081
256	69 391	1532	1522	510	1024	20.934
512	275 267	3068	3058	1022	2048	57.337
1024	1 096 505	6140	6130	2046	4096	115.382

“Coordinate bits” is the sum of the binary coordinate lengths, with one bit charged for zero; it is not the maximum coordinate length. “Surface” and “Prism” count face-pairing bands, “Contacts” counts the exceptional prism-to-residue contacts, and “Residue” counts local residual cells. At $t = 128$ the input represents exactly

$$2\,791\,715\,456\,571\,051\,233\,611\,642\,548$$

normal discs, but only 764 surface bands and 754 prism bands are produced. For two fixed tetrahedra with 65,536-bit multiplicities, there remain ten stacks, four surface bands, and four prism bands; the recorded seven-run median is 0.000600 seconds. These are representation and extraction measurements. They do not include general orbit reduction, parallelity bundle recognition, or a complete hierarchy iteration.

9.5 Attached copies: exact equation savings and a crossover

The attached-copy benchmark compares the modified backend with the pinned `scalar_split.py`; both use the same remaining dependency modules. The measured region is recursive scalar compression followed by component sharing. Fixture construction is outside the timed region.

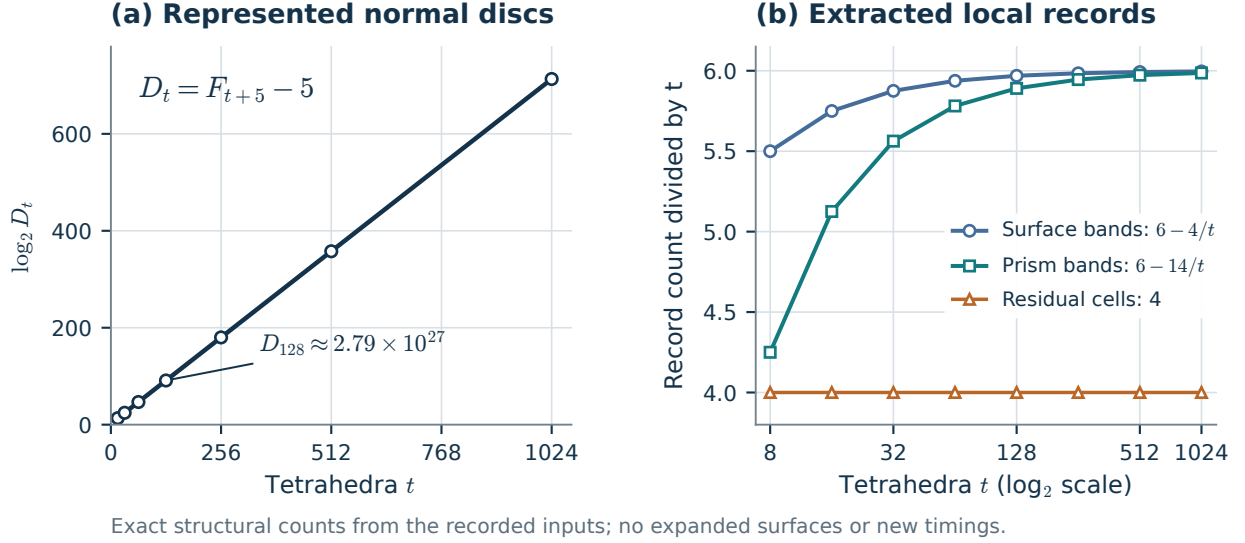


Figure 2: Exact structural counts for the inherited Fibonacci meridian family, at the recorded values $t = 8, \dots, 1024$. The normal-disc count is $D_t = F_{t+5} - 5$. Its binary logarithm grows linearly with t in panel (a), while panel (b) shows bounded numbers of extracted records per tetrahedron. The horizontal axes use different scales, as labelled. These are count data, with no fitted running-time exponent.

r	Old equations	Inherited	Old solves	New	Ratio	A/A
4	58	32	3	1	0.848	1.027
8	406	128	7	1	1.099	1.051
12	1 298	288	11	1	1.252	1.013
16	2 990	512	15	1	1.468	0.999
24	9 798	1 152	23	1	1.759	0.989

The equation counts are the exact values proved in Section 7. At $r = 16$, repeated solves construct 2,990 nonzero equations across fifteen commutant computations, while inheritance constructs 512 equations in one computation. Candidate order and accepted split witnesses agree. The 1.468 paired ratio reflects an actual measured benefit. At $r = 4$, the ratio 0.848 instead shows the cost of transport and canonicalization exceeding the saving.

The $r = 24$ experiment raises the backend caps: its 72 objects and 1,728 scalar variables exceed the default limits, and its 23 splits exceed the default split budget of sixteen. It is a controlled scaling experiment, not a claim about behavior under default caps. The $r \leq 16$ rows fit the default object and variable limits. Neither the exact cubic-to-quadratic equation reduction nor the measured ratios establish an analogous total bit-time exponent for arbitrary complexes.

9.6 Field algebras: identify the work that actually matters

For the dense two-field family, the reference is the previously supplied primary-projector policy, with descendant commutants recomputed. This is a separate reference from the pinned zero-primary Fitting search used for the preceding table. The new mechanisms are measured independently:

r	Inheritance	Local stop	Both	One generator	A/A
8	0.890	1.877	1.446	2.165	1.003
10	0.868	2.033	1.565	2.312	1.002
12	0.895	1.935	1.489	2.229	0.985

The four variant columns mean general corner inheritance only, direct saturation-locality stopping only, their ordinary combination, and the one-generator corner path with locality certification. General inheritance alone is slower in all three displayed cases. Carrying the full basis is not the right optimization for these low-dimensional commutative algebras. The saturated generator exposes the complete algebra and finishes both local children directly.

For degrees $r = 8, 10, 12$, the old candidate counts are respectively 46, 52, and 58. The one-generator path uses two candidate tests and one commutant computation, followed by two scalar-locality certificates. The reference uses three commutant computations. The corresponding paired ratios are 2.165, 2.312, and 2.229. The $r = 12$ root has 1,152 scalar variables, exceeding the default variable cap, although its 48 objects fit the default object cap.

The data also retain lower-degree cases and a degree-eight case using only one elementary change of basis to mix the two blocks. Not every accepted generator saturates the algebra; the latter case falls back to ordinary inheritance, and direct local stopping is faster. This is evidence for a cost-aware choice of mechanism, rather than for turning every optional feature on unconditionally. All these fixtures are legitimate homogeneous algebraic complexes; no realization as natural knot scan prefixes is claimed.

9.7 Natural scans: a useful negative timing result

Natural examples measure a complete raw Khovanov scan, including scanner construction, at the same preselected order for every arm. Loading the input and finding that order are excluded. Homological ranks and the configured capped decisions are checked independently. The timing table is:

Diagram	Crossings	Old ms	Inherit	One gen.	A/A	Solves
Trefoil	3	0.319	0.961	0.958	1.003	$0 \rightarrow 0$
Conway	11	7.530	0.990	1.018	0.954	$10 \rightarrow 7$
Kinoshita–Terasaka	11	8.079	1.022	1.010	1.016	$9 \rightarrow 6$
Hard unknot 8	8	0.732	1.001	0.993	1.002	$0 \rightarrow 0$
Stress braid 5/36	36	457.241	1.021	0.980	1.005	$5 \rightarrow 5$
$T(3, 5)$	10	2.917	0.998	0.994	0.982	$0 \rightarrow 0$

The final column counts actual commutant computations; for the historical arm, cache misses equal those computations in these cases. Conway falls from ten to seven and Kinoshita–Terasaka from nine to six, each using three inherited child spaces across two Fitting splits. There is a real reduction in repeated algebra construction. Wall-time inheritance ratios, however, range only from 0.961 to 1.022, comparable with the control variability. These measurements support no convincing natural-knot wall-time speedup. No natural example in this sample activates a primary split, a conclusive locality certificate, or a monogenic corner. The controlled field gains must remain separate from claims about the complete recognizer.

9.8 Integrated regression and the inherited worker correction

The untouched pinned baseline ran 725 tests: 724 passed and one failed. The failure in `test_communication_retains_input_across_polls` was reproduced in isolation. Repeating `communicate` with the input removed after a timeout could leave a large worker request only partially written in the current Python runtime. The earlier projectors archive already contained a single-submission communication-thread correction. We copied that correction exactly and attribute it as inherited work. It is not counted as a new performance theorem or optimization.

The complete staged source bundle then passed **790 tests**, with no failures, in 110.342 seconds of reported test-runner time. This includes all 725 baseline methods, twelve restored tests for the inherited primary projector module, and 53 new arithmetic, family, normal-extraction, and inheritance/locality methods. Targeted scalar checks also preserve cache, cap, cancellation, grading, and transactional behavior. The logs are included, including the original baseline failure; the differing total suite times are not used as a performance comparison.

10 Further research questions and prioritized experiments

The most useful next work should connect the proved interfaces to genuine residual knot instances. The following questions are proposed targets, not claims that each formulation is absent from the literature.

10.1 From normal coordinates to a complete compressed cut

1. **Complete the residual face attachment map.** The current extractor identifies local non-prism cells and their prism contacts. Construct all residual faces, edges, and their identifications, with an independent expanded oracle on small normal surfaces. The acceptance criterion is a complete cut manifold representation, not only correct piece counts.
2. **Connect a general interval-orbit implementation.** Concatenate the signed bands into the integer-interval format of Agol–Hass–Thurston and compute connected components without expansion. Then add weighted orbit data for Euler characteristic, boundary count, and orientability. An existing independent orbit implementation can be used as a comparison oracle, with its provenance retained.
3. **Certify cyclic blocks inside interval data.** Find conditions under which a family of partial interval maps is equivalent to total translations on a common cyclic fibre. Return the equivalence and attachment transport as a checkable witness. This is the missing bridge between the two geometric kernels in this article.
4. **Follow actual nonextreme normal surfaces.** The earlier primitive-extreme-ray disc checker deliberately avoids component enumeration. Test the new extractor on connected and disconnected nonextreme sums, one-sided surfaces, boundary-parallel surfaces, and normal surfaces with large quadrilateral coordinates. Those examples exercise cases not certified by primitivity alone.
5. **Track boundary patterns through compression.** Extend local provenance to pattern arcs and vertical annuli. A useful milestone is one complete verified iteration of the geometric hierarchy on a knot exterior, including the boundary-pattern update and a replayable certificate.

10.2 Stronger symbolic queries with controlled expressiveness

6. **Geometrically restricted peripheral constraints.** The three-colouring reduction shows that arbitrary simultaneous peripheral connectedness is too expressive for an unrestricted polynomial claim. Identify geometric restrictions, such as bounded incidence width or a fixed number of designated boundaries, that yield tractable subclasses.
7. **Fixed-sign dihedral families.** If reflection signs are fixed and only shifts vary affinely, determine which component-minimization questions reduce to a bounded number of affine-gcd problems. Treat parity, fixed residues, and the small moduli 1, 2 explicitly.
8. **Joint objectives for disconnected bases.** The two-component counterexample shows that independent minima cannot simply be added. Study fixed-parameter algorithms in the number of base components or in the incidence graph between parameter variables and component objectives.
9. **Boundary-profile existence.** Seek algorithms for one or a fixed number of prescribed boundary-cycle profiles. Distinguish a bounded disjunction of congruence systems from a claim that all possible genus and boundary constraints are polynomial.
10. **Counting with supplied arithmetic structure.** Given a factorization of m , implement the exact prime-local count formula and compare it with enumeration on small cases. Without a factorization, study partial counts or certificates that do not implicitly solve the totient problem.

10.3 Scalar compression and actual scanner activation

11. **Measure natural commutant reuse.** Record parent algebra dimension, actual child dimensions, equation counts, transport costs, and candidate counts on difficult prime knots that survive early filters. A controlled algebraic speedup should not be reported as a complete-recognizer improvement without this measurement.
12. **Discover saturated generators efficiently.** A generator Q with $\dim \mathbb{F}_2[Q] = \dim \text{End}(C)$ supplies a complete commutative description. Study deterministic candidate construction and characterize which scalar algebras admit such a generator. Noncommutative or nonmonogenic algebras require a different certificate.
13. **Certificates for larger nonsplit algebras.** Generalize the locality certificate to noncommutative finite-dimensional scalar algebras using their radical and semisimple quotient. Charge both the cost of obtaining the algebra and the cost of the certificate.
14. **Continuation-compatible inheritance.** Determine which parts of an endomorphism-space description survive the next crossing attachment. Arbitrary reuse across changed differentials is unsound; the update must prove the relevant intertwining conditions.
15. **Formalize the small arithmetic core.** The affine merge lemma, bounded-residue congruence elimination, and dual verifier are suitable for a compact Lean formalization. The proof should include the composite-modulus case and the exact meaning of the bit-cost model.

10.4 A credible route to the global target

16. **Prove an amortized hierarchy reset bound.** A natural target is a quantity whose total loss bounds all failed repairs and is only quasi-polynomial in the original diagram size. Merely restating a lexicographic termination proof is not enough.
17. **Bound the encoding of every reachable state.** Include normal weights, boundary patterns, attachment maps, coefficient sizes, and any family parameters. A finite number of local topological types does not by itself bound the multiplicity or marking data.
18. **Find a complete residual family with proven coverage.** Choose an explicit infinite class that defeats inexpensive existing certificates, and prove that the remaining geometry always enters the implemented compressed model. This would be more informative than a large synthetic speedup alone, even if the resulting theorem remains restricted to that class.

11 Reproducibility and repository integration

11.1 What the archive contains

The archive includes the complete article source, generated tables and scientific figures, and the compiled PDF. A curated `source/` tree contains the maintained Python implementation, its tests and fixtures, and the historical reference modules required by those tests. Historical result archives unrelated to this continuation are omitted. The distributed source is pinned to `ea9cf443c8ce` except for the explicit delta in `source_manifest.json`.

The `patches/` directory contains a Git-applicable patch with two modified files and fourteen new files. A separate `baseline/` directory retains the two modified files exactly as they appeared at the pinned revision. It lets the scalar benchmark load its historical arm without network access or a Git checkout. Independent validation applied the patch in a fresh directory, checked all sixteen affected outputs byte for byte, and compared every one of the 362 distributed source files with the working tree. The 346 unmodified source files match the pinned Git blobs exactly.

The `experiments/` directory contains the arithmetic and geometric reference checks, including rational geometry and optional Regina comparisons. `results/` preserves the recorded measurements and exact comparison counts. `validation/` preserves the baseline failure, final 790-test success, targeted checks, patch audit, and artifact checks. Standalone copies of the small arithmetic and extraction modules are kept only to make those experiment scripts independent of package installation; their bytes match the corresponding integrated modules.

11.2 Commands for independent reproduction

The new mathematical kernels require only the Python standard library. Regina is optional for the independent native audit; Matplotlib is optional for regenerating the figures. The recorded versions are listed in `requirements-audit.txt`. LaTeX compilation uses ordinary `pdflatex` and `latexmk`, with the packages declared in the main source file.

From the extracted archive root:

```
python scripts/check_hashes.py
python scripts/reproduce.py quick

# Optional native audit and figure dependencies:
python -m pip install -r requirements-audit.txt
python scripts/reproduce.py full

# Paired and binary-size benchmarks:
python scripts/reproduce.py benchmarks

# Rebuild the article:
cd article
latexmk -pdf -interaction=nonstopmode -halt-on-error \
    unknot_affine_families.tex
```

The reproduction driver writes new logs and measurements to `reproduction/`; it preserves the recorded evidence. “Quick” runs all 65 methods in the five new or restored test modules, the rational face-order audit, the certificate examples, and the peripheral reduction check. “Full” runs the complete integrated suite and the Regina audit. The benchmark mode is separate so that correctness checks do not depend on machine-specific timing thresholds.

To review the source delta at the pinned repository revision, use:

```
git apply --check /path/to/affine_families_and_inheritance.patch
git apply /path/to/affine_families_and_inheritance.patch
```

The article and supporting research files can then be placed in the repository’s chosen report or incoming-document location. The distributed patch edits the implementation paths; it does not choose or overwrite a report number.

11.3 Concrete example contracts

The cyclic-family examples include a constrained annulus with optimum two, a self-seam joining twelve locally disconnected annuli into a connected member, an infeasible seam with a negative certificate, and a 10,589-bit modulus whose stored hexadecimal witness verifies without sheet expansion. The K_4 example keeps its designated-boundary requirements in a separate metadata file, since that hard peripheral query is outside the optimizer’s strict input schema. Run:

```
python scripts/check_family_examples.py
```

For direct package use, the public family functions are `compile_family`, `optimize_cover_family`, `realize_family`, and `verify_cover_family_certificate` in `fastunknot.cyclic_cover_family`. The command-line entry point accepts one JSON input and emits the compiled family and arithmetic certificate. The verifier ignores the producer’s compilation and recomputes it from the original geometry. It certifies feasibility and optimality, or infeasibility; it does not certify the auxiliary assignment count or any topological predicate outside the stated component objective.

The normal extractor accepts a JSON object with exactly `triangulation` and `coordinates`. It emits the signed polygon bands, prism bands, residual-cell incidence data, and provenance. The optional `--orbit-inputsurface` and `--orbit-inputorientation` switches emit the corresponding interval orbit problems. Those files are inputs to a subsequent orbit engine, not claims that the engine has been run. The supplied normal examples include the sharp two-tetrahedron fixture and

a 128-tetrahedron Fibonacci meridian.

11.4 Optional scalar controls and evidence boundaries

Within the already optional Fitting backend, ordinary reuse defaults to `fitting_reuse_endomorphisms=True`. It preserves the deterministic candidate basis and search policy. The separately configurable primary, locality, and one-generator paths default to false:

```
from fastunknot.scalar_split import fitting_khovanov_rank
answer = fitting_khovanov_rank(
    pd,
    fitting_reuse_endomorphisms=True,
    fitting_primary=True,
    fitting_certify_local=True,
    fitting_monogenic_corners=True,
)
```

This snippet deliberately selects an experimental backend and its optional mechanisms. It is not a recommendation to raise every size cap on arbitrary inputs. Independent locality auditing uses `verify_scalar_local_certificate`; unlike production inheritance, that verifier recomputes the full commutant to check the dimension claim. The different cost contracts of the arithmetic and scalar verifiers should not be conflated.

11.5 Provenance and continuing obligations

The inherited polynomial-projector module, its fixtures and tests, and the worker-communication correction are attributed by source path and hash. The Fibonacci meridian family is likewise attributed to the earlier compressed-certificate report. No external article PDF or downloaded slide deck is bundled as a new research artifact. The new work follows the repository’s MIT No Attribution license; included pre-existing components retain their accompanying license notices.

A claim ledger records proved statements, tested implementation contracts, measured improvements, negative results, and unresolved global obligations. The immediate deliverable is a set of exact reusable operations and an integration-ready research record. A complete quasi-polynomial recognizer still requires the explicit state-growth, representation-size, and geometric-coverage arguments in Section 8.

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